

DSA STUDY BLUEPRINT SERIES

66. Design a Min Stack Optimal Solution

Designing a Stack supporting $O(1)$ Minimum lookup

Section 08 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Min Stack:** Supports push, pop, top, and getMin in constant time.
- Keep an auxiliary stack that tracks min values, or encode values.

Memory & Execution Blueprint

Main Stack:
[5, 3, 7]

Min Stack:
[5, 3]

C++ Implementation Reference

Standard code layout and syntax

```
class MinStack {  
    stack st, minSt;  
public:  
    void push(int val) {  
        st.push(val);  
        if (minSt.empty() || val <= minSt.top()) minSt.push(val);  
    }  
    void pop() {  
        if (st.top() == minSt.top()) minSt.pop();  
        st.pop();  
    }  
    int top() { return st.top(); }  
    int getMin() { return minSt.top(); }  
};
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
All Operations	O(1) time, O(N) space

Key Practices & Gotchas

- **Important:** Use inequality checks (`val <= minSt.top()`) to handle duplicate minimums correctly.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace monotonic stack modifications for next greater element on `[4, 5, 2]` :

Active Element	Stack State (Indices)	Pop comparison	Result output
4	[0] (value 4)	N/A	Waiting for next greater
5	[1] (value 5)	$5 > 4$ (pop index 0)	Next Greater of 4 is 5

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Monotonic properties:** Stacks process elements strictly increasing or decreasing, optimizing nested lookups.
- **Min Stack details:** Track current minimum bounds by storing elements as encoded difference values.
- **Related LeetCode Problems:** #155 Min Stack, #84 Largest Rectangle in Histogram, #42 Trapping Rain Water.